This project aims to identify key sonar frequency patterns (45–260 kHz) that differentiate Lake Trout from Smallmouth Bass for classification.

Step 1: Feature Selection

For the first step of my exploratory data analysis, I focused on feature selection by creating two filtered datasets based on different analytical needs:

- fish_data: Contains fish label (fishNum), species (species), biological features (total length, fork length, weight, girth, dorso-lateral height, sex, air bladder total length, air bladder width, and air bladder weight), and frequency response (F45–F260).
    - Purpose: To analyze the relationship between biological features and frequency responses across species.
- freq_only: Contains only fish label (fishNum), species (species), and frequency response (F45–F260).
    - Purpose: To examine frequency response patterns independently of biological features.

Before further analysis, I addressed missing data to ensure dataset completeness and reliability. The next step involves identifying and handling missing data to improve data quality.

Step 2: Cleaning Data

To ensure data reliability, I removed fish with excessive missing values or insufficient observations. I examined the number of pings per fish and determined that fish with fewer than 40 pings provided insufficient data for meaningful frequency response analysis. These fish were excluded to prevent potential biases caused by limited observations. I visualized missing data using bar plots and heatmaps, which highlighted frequency gaps across individual fish.

Lake Trout 008 (LT008) was removed as it had over 50% missing values in the 173–260 kHz range, exceeding the acceptable threshold for analysis. Additionally, LT010, LT013, LT015, SMB009, and SMB011 had low ping counts, making their frequency response data unreliable. To improve consistency and data quality, I also removed the 90–170 kHz frequency range, as it exhibited excessive noise and inconsistencies. These cleaning steps ensured that the remaining dataset was complete, consistent, and suitable for species classification and feature selection.

Step 3: Frequency Response Pattern Analysis

With the cleaned dataset, I now analyzed species-specific frequency patterns to identify distinguishing features for classification.

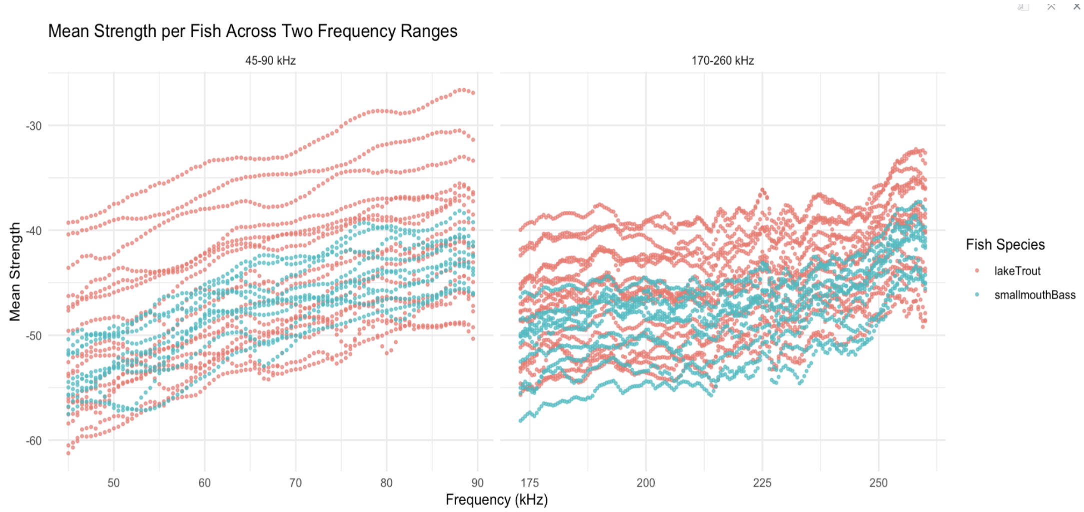

Figure 1.

Figure 1 shows the mean strength per fish across 45–90 kHz and 173–260 kHz. This visualization enables direct species comparison. The analysis reveals that Lake Trout exhibits stronger frequency responses than Smallmouth Bass across both ranges. However, species separation is clearer in the 45-90 kHz range, suggesting that lower frequencies might be more informative for classification. To further evaluate these differences, I conducted independent t-tests at each frequency, adjusting for multiple comparisons using Bonferroni correction.

Step 4: Selecting Key Frequencies for Species Classification

To optimize classification, I identified the frequencies that most effectively differentiate species.

Frequency_kHz <dbl>	p_value <dbl>	t_statistic <dbl>	p_adjusted <dbl>
54.5	7.482594e-54	15.58231	1.982887e-51
54.0	4.876040e-53	15.45717	1.292151e-50
53.5	4.464997e-52	15.30635	1.183224e-49
53.0	9.769373e-52	15.25353	2.588884e-49
55.0	1.499341e-51	15.22502	3.973255e-49
52.5	2.838283e-51	15.18023	7.521449e-49
52.0	3.064370e-48	14.70058	8.120581e-46
51.5	1.461011e-46	14.42937	3.871680e-44
55.5	2.425124e-46	14.39219	6.426577e-44
84.5	8.776901e-41	13.45834	2.325879e-38

Table 1.

Table 1 summarizes the t-test results, highlighting the most significant frequency bands where Lake Trout and Smallmouth Bass exhibit statistically distinct signal strength patterns.

The results in Table 1 show that the 51.5–55.5 kHz range provides the strongest species separation, with extremely low adjusted p-values (p_adjusted), confirming high statistical significance. Additionally, the consistently positive t-statistics indicate that Lake Trout exhibits higher strength values than Smallmouth Bass in this range. These findings suggest that the 51.5–55.5 kHz range is highly informative for species classification and should be prioritized in predictive modeling.

I also explored Principal Component Analysis (PCA), which identifies frequencies that contribute the most to overall variance in the dataset. PCA selected the 63–77 kHz range as the most important for explaining total variance, but these frequencies did not effectively differentiate species. Thus, I selected the t-test-identified 51.5–55.5 kHz range for further analysis, as it provided the clearest species separation. While PCA is valuable for dimensionality reduction, it was not the best method for species differentiation in this dataset. This comparison highlights the importance of testing multiple methods in exploratory analysis before selecting the most appropriate features for classification.

Step 5. Correlation Analysis

Next, I examined correlations among frequencies to identify redundant signals and assessed how biological features influence frequency response

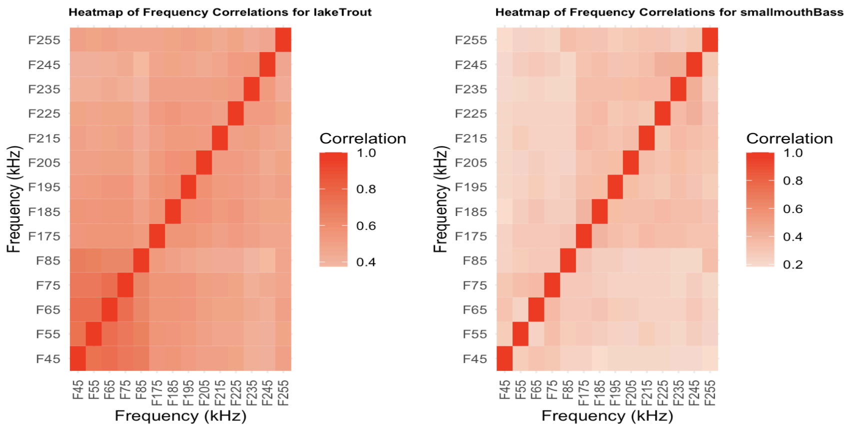

Figure 3.

To assess redundancy in frequency responses, I generated species-specific correlation heatmaps (Figure 3). These show that adjacent frequencies exhibit strong correlations, meaning that some frequency bands might be redundant. Lake Trout responses appear more stable across frequencies, while Smallmouth Bass exhibit greater variation, especially at higher frequencies.

	species	Variable1	Max_Correlation	Mean_Correlation	Frequency_Count	Max_Correlation_Frequency
1	lakeTrout	airBladderWidth	0.5970510	0.3841084	75	F67
2	lakeTrout	airbladderWeight	0.3169670	0.2868557	0	F86.5
3	lakeTrout	weight	0.3068690	0.2650368	0	F214
4	lakeTrout	girth	0.3047476	0.2556438	0	F86
5	lakeTrout	forkLength	0.2896842	0.2448220	0	F214
6	lakeTrout	totalLength	0.2772064	0.2308163	0	F214
7	lakeTrout	airbladderTotalLength	0.2164455	0.1835817	0	F218
8	smallmouthBass	girth	0.5538209	0.4150477	13	F174
9	smallmouthBass	weight	0.5495564	0.4096372	10	F174
10	smallmouthBass	forkLength	0.5468301	0.4067978	10	F174
11	smallmouthBass	totalLength	0.5462755	0.4059560	10	F174
12	smallmouthBass	airbladderWeight	0.5129498	0.3673921	2	F174
13	smallmouthBass	airbladderTotalLength	0.4874739	0.3583062	0	F174
14	smallmouthBass	airBladderWidth	0.4308073	0.3181654	0	F174

Table 2.

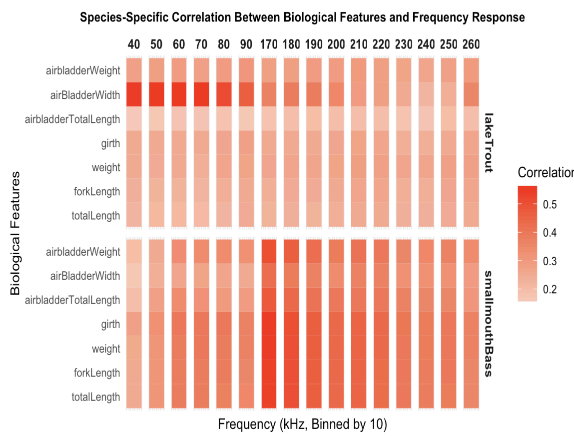

Figure 4.

To analyze the relationship between biological features and frequency response, I used both a heatmap (Figure 4) and a summary table (Table 2) for Lake Trout and Smallmouth Bass. These visualizations provide complementary perspectives: the heatmap highlights broad correlation trends across frequencies, while the table ranks biological features by their maximum correlation and identifies the frequency at which the correlation peaks. Together, these approaches help determine whether species differentiation is influenced by a single dominant feature or a combination of multiple moderately correlated features.

The results reveal species-specific patterns. For Lake Trout, air bladder width exhibits the highest correlation, particularly around F67, suggesting its key role in shaping frequency response. In contrast, for Smallmouth Bass, girth and weight show the strongest correlations, particularly near F174, indicating that body size plays a larger role in its acoustic signature. The heatmap allows for broad pattern recognition, while the table provides a structured ranking of key features, making both useful for feature selection. These results highlight distinct species-specific frequency correlations, supporting the need for tailored classification models.

Step 6. Sex-Based Comparison

I next examined how sex influences biological features, as morphological differences could affect frequency response patterns and impact species classification models.

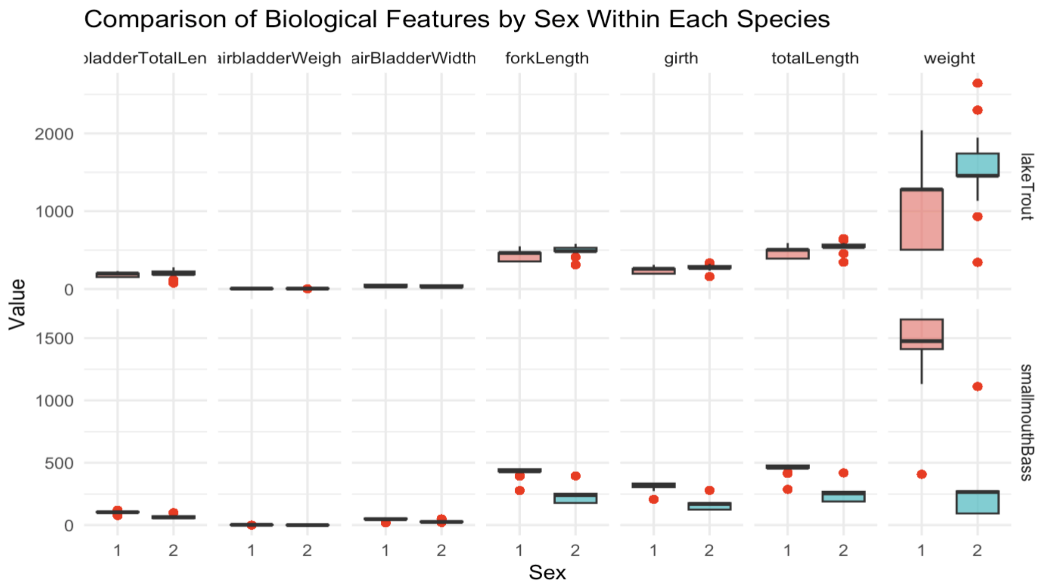

Figure 5.

Figure 5 compares biological features across sexes for Lake Trout and Smallmouth Bass.

For Lake Trout, males and females show similar distributions for most features, though female weight exhibits greater variability. In contrast, Smallmouth Bass males tend to have higher girth and weight, indicating more pronounced sex-based differences. Some outliers are present in both species, particularly for weight and total length, suggesting the presence of unusually large or small individuals. These findings suggest that while sex may influence some biological features, its impact on classification models should be carefully evaluated to determine its relevance to frequency response patterns.